

ME 300 Homework 6: Steady-Flow Energy Equation and Steady-Flow Devices

Answer the following questions neatly on a separate piece of paper or on a tablet. Written answers can be typed. Show all your work, list all your assumptions, and explain your reasoning – partial credit will only be given if you show and explain each of your steps. Submit your homework on Canvas.

1. (10 points) Describe flow work in your own words. Use the analogy from the video to help support your discussion.
2. (20 points) A variable-area nozzle is used to accelerate steady-flowing air ($c_p=1001 \text{ J/kg-K}$) to different flow velocities. The air always enters the nozzle at a velocity of 10 m/s, a temperature of 350 K, and density of 1.225 kg/m^3 , where the nozzle has an initial area of 0.02 m^2 .
 - a. What is the mass flow of air through the nozzle?
 - b. Plot the temperature of the air leaving the nozzle as a function of nozzle exit velocity from 20-200 m/s. Show your calculation steps on your homework paper and then use Excel or Matlab to do the calculations at all the points requested.
3. (15 points) A throttle is used to reduce the pressure of steady-flowing water in a processing plant. The water enters the throttle at a pressure of 5 MPa as a saturated vapor and leaves the throttle at a pressure of 3 MPa. What is the temperature and mass-specific internal energy of the water as it leaves the throttle?
4. (25 points) In an experiment in my lab we use hot air to vaporize water before putting it into the combustor (this is real and we really did this calculation earlier this year!). Treat this problem like a two-fluid heat exchanger, where the air ($c_p=1001 \text{ J/kg-K}$) enters the heat exchanger at 623 K and the water enters the heat exchanger as a saturated liquid at 0.1 MPa. Use an air flow rate of 0.05 kg/s and the water flow rate is 0.002 kg/s. What is the temperature of the air leaving the heat exchanger if the water leaves as a saturated vapor? How much heat was transferred? Make sure to list all assumptions you used.
5. (25 points) The gas turbine in University Park's West Campus Steam plant runs constantly to provide campus with power and steam. We've collected data on the engine's compressor for a few days in June – see the data sheet attached in the Homework 6 assignment in Canvas. Use this data sheet to calculate the work done by the compressor to compress air in the engine. Assume a mass flow rate of 30 kg/s and the following values for air: $R=287 \text{ J/kg-K}$ and $c_v=714 \text{ J/kg-K}$.
 - a. List all your assumptions on your paper as well as the method you're going to use to calculate these values in the spreadsheet.
 - b. Plot the work done by the compressor as a function of time. BE CAREFUL ABOUT UNITS – the units on the spreadsheet are not the ones you should use in the calculation.
 - c. Discuss this trend – what's changing with time and why do you think that is? Do some research on how a gas turbine works – what role does the compressor play in the machine? List your sources for where you found info on gas turbines.
6. (5 points) Fill out the course feedback survey on Canvas – we've had all the elements of the course so far (lectures, homework, quizzes, explained examples, and an exam). Please provide feedback on how the course is going so we can re-calibrate for the remainder of the summer if changes are needed.